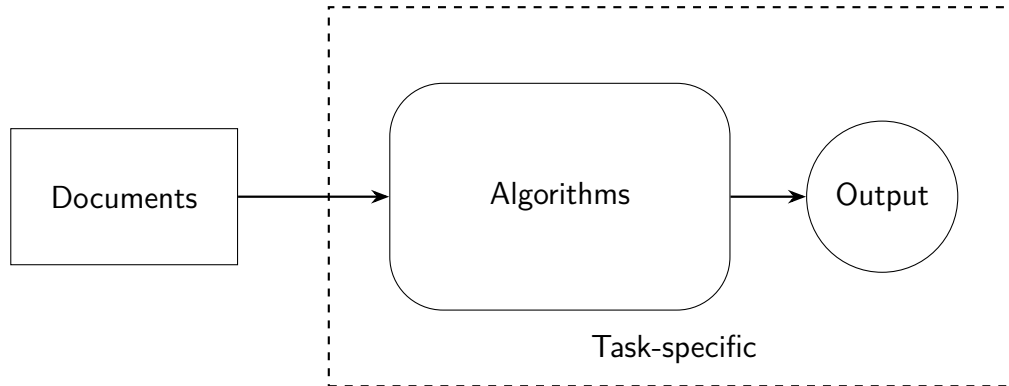


1 Lecture 2 (Representing Text)

1.1 Input Representation



Definition 1.1 (Raw Term Frequency / Term count): The raw count is the number of times the word t appears in the document d , denoted by $c(t, d)$.

Definition 1.2 (Term Frequency):

$$tf(t, d) = \begin{cases} 1 + \log_{10} c(t, d), & \text{if } c(t, d) > 0 \\ 0, & \text{otherwise} \end{cases}$$

Definition 1.3 (Document Frequency): The number of documents containing the word t , denoted by $df(t)$.

Definition 1.4 (Inverse Document Frequency):

$$idf(t) = \log_{10} \frac{N}{df(t)},$$

where N is the total number of documents in the corpus and $df(t)$ is the document frequency.
More common form used in practice,

$$idf(t) = \log_{10} \frac{N}{1 + df(t)},$$

or

$$idf(t) = \log_{10} \left(1 + \frac{N}{df(t)} \right)$$

NLP model Representations

Statistical idea SVD Truncated SVD Term document matrix

1.2 Latent Dirichlet Allocation (LDA)

LDA generative process, example corpus

Algorithm 1 Generative Process (Latent Dirichlet Distribution)

```

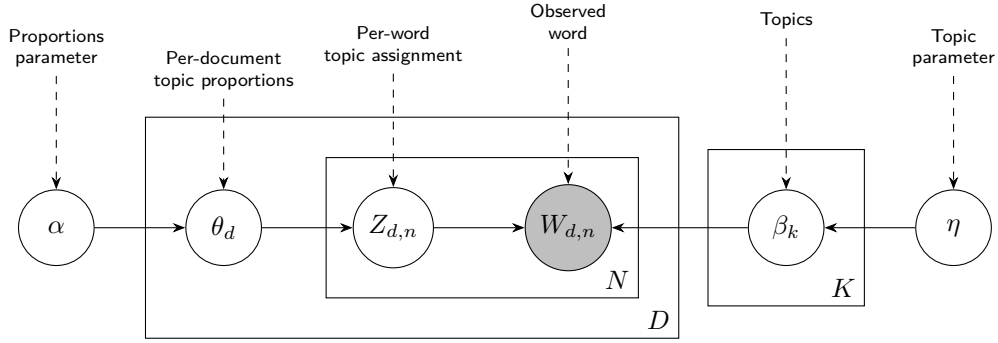
1: for topic  $k \in \{1, \dots, K\}$  do
2:    $\phi_k \sim \text{Dir}(\beta)$  [draw distribution over words]
3: end for
4: for document  $m \in \{1, \dots, M\}$  do
5:    $\theta_m \sim \text{Dir}(\alpha)$  [draw distribution over topics]
6:   for word  $n \in \{1, \dots, N_m\}$  do
7:      $z_{mn} \sim \text{Mult}(1, \theta_m)$  [draw topic assignment]
8:      $x_{mn} \sim \phi_{z_{mn}}$ 
9:   end for
10: end for

```

Remark 1.5: ϕ_k is the word topic distribution, and θ_m is the document topic distribution.

Definition 1.6 (Dirichlet Distribution):

$$p(\theta \mid \alpha) = \frac{\Gamma(\sum_{i=1}^k \alpha_i)}{\prod_{i=1}^k \Gamma(\alpha_i)} \prod_{i=1}^k \theta_i^{\alpha_i - 1}$$



$$p(\beta, \theta, z, w) = \left(\prod_{i=1}^K p(\beta_i \mid \eta) \right) \left(\prod_{i=1}^K p(\theta_d \mid \alpha) \prod_{n=1}^N p(z_{d,n} \mid \theta_d) p(w_{d,n} \mid \beta_{1:K}, z_{d,n}) \right)$$

1.2.1 Parameter estimation using Gibbs Sampling

$$p(z \mid w, \alpha, \beta) = \frac{p(z, w \mid \alpha, \beta)}{\sum_z p(z, w \mid \alpha, \beta)}$$

$$P(z_i = j \mid \mathbf{z}_{-i}, \mathbf{w}) \propto \frac{n_{-i,j}^{(w_i)} + \beta}{n_{-i,j}^{(\cdot)} + W\beta} \frac{n_{-i,j}^{(d_i)} + \alpha}{n_{-i,\cdot}^{(d_i)} + T\alpha}$$

Supervised Topic Models

sLDA

heirarchial topic model correlated topic model

Word Representations

2 Lecture 3 (Word Representations)

2.1 Words as vectors: Word2Vec

Representation of a word is dictated by other surrounding Words Assume a fixed length context window. Eg: $[w_{-1} \ w_{-1} \ c \ w_1 \ w_2]$

Start with random initialization Iterate till convergence

2.2 Word2Vec Model: SkipGram(SG)

2.3 Word2Vec Model: CBOW

2.4 Objective function

$$f_p = \frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c} \log p(w_{t+j} | w_t)$$

$$p(w_o | w_1) = \frac{\exp(\text{dot}(v'_{w_o}, v_{w_1}))}{\sum_{w=1}^W \exp(\text{dot}(v'_w, v_{w_1}))}$$

With negative sampling, the objective function becomes

$$\log \sigma(v'_{w_0} \cdot v_{w_l}) + \sum_{i=1}^K \log \sigma(-v'_{w_i} \cdot v_{w_l})$$